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Methods, Systems and Computer Program Products for Handling Data Records Using an Application Programming Interface (API) and Directory Management System

Abstract

An automated computer implemented method is provided to receive images and metadata from an acquisition device; parse the received metadata into data entities based on a standardized ontology; automatically write a data acquisition record into an acquisition side database that includes protected information; format the received images into a standardized image format, the image format including a set of information necessary to visualize and quantify the image; store the formatted images on a recording medium; populate a storage record into a storage side database that is free of protected information; and link information on the acquisition side database and the storage side database using a unique code.

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Background/Summary

CLAIM OF PRIORITY [0001] The present application is a continuation of U.S. patent application Ser. No. 17/870,317, filed on Jul. 21, 2022, which claims the benefit of and priority to U.S. Provisional Application Ser. No. 63/224,503, filed on Jul. 22, 2021, the contents of which are hereby incorporated herein by reference as if set forth in their entireties.

BACKGROUND

[0003] The management of medical images and data in the context of research and clinical trials is notoriously difficult even as the opportunities and demand for imaging biomarkers and artificial intelligence clinical decision support systems is rapidly expanding. Successful development, validation, and clinical deployment of imaging biomarkers requires vast amounts of images supported by accurate clinical annotations. In the domain of artificial intelligent (AI) and big data, there is an urgency to deploy image and data management platforms that are Findable, Accessible, Interoperable, and Reusable (FAIR). This gets complicated in a multi- vendor, multi-modal, multi-site environment that lacks unified standards on imaging systems, metadata, and observational records.

[0004] A successful image management platform that addresses the needs of imaging biomarker development will be quite differentiated from a more traditional medical imaging Picture Archiving and Communications System (PACS). PACS systems are designed around access and visualization of images under well-defined conditions and are most suited to clinical diagnostic applications. While PACS system development has made significant advances in network availability and visualization, integration with observational medical records remains a challenge, and PACS systems are generally not suited to the management and analysis of populations of images in the big data and AI context.

[0005] A successful platform for the collection and utilization of images in the modern context must address the needs of diverse stakeholders, from research, to clinic, to patient. The unstructured, semi-structured, and structured data that makes up the medical imaging and health record environment must be easily findable by administrators and researchers alike, with appropriate security and respect of individual privacy rights. The data must generally be Accessible at the granularity required for specific tasks, and in such time frames as appropriate to various image-driven workflows. The images must be Interoperable across vendors, and appropriately interoperable across interrelated, correlated, or causal modalities. Images and data must generally then be reusable. Data definitions must generally be standard enough to be reused from study to study in the capture of new data, specific enough capture details required in complex medical domains, broad enough to be extensible to new domains of inquiry.

[0006] Clinical trials require aggregation of volumes of data from a multiplicity of examination devices, acquired on a large cross section of patients at often large number of clinical trial sites. Data quality, traceability, and security are essential in the collection, transport, and analysis of clinical trial data.

[0007] Clinical trials rely on combinations of biomarkers and endpoints to validate therapeutic

safety and efficacy before regulatory bodies. The onus of developing robust, objective clinical trial endpoints places additional demands on the data management. Clinical trial endpoints are often developed in parallel with the pharmaceutical development and clinical trial process. Novel endpoints may be exploratory, without direct bearing on a regulatory decision, or secondary, as an adjunct to a primary definitive endpoint. In general, regulatory bodies seek structural endpoints, functional endpoints, and endpoints that directly tie to activities of daily living.

[0008] As the eventual primary or secondary endpoints may evolve through development and clinical trial process, it is particularly important to institute a robust data management process early, and systematically collect and organize exam data and exam metadata from the beginning, to maximize the usability to answer new questions that arise through the development lifecycle.

SUMMARY

[0009] Some embodiments of the present inventive concept provide an application programming interface (API) resident on a computer that, upon receiving a message that a data set associated with an encounter experienced by a subject is available, causes a processor to: read a structured file associated with the data set, the structured file having a set of records assigned to a subset of pre-defined fields and a location of one or more unstructured objects, the pre-defined fields being organized according to an ontology having at least records related to the subject, the encounter and the data set; and populate at least first and second independent and distinct relational databases using endpoints of the API, the first relational database including records related to the one or more unstructured objects and excluding records associated with the subject and the encounter and the second relational database including records associated with the subject and the encounter. A schema of the second relational database includes at least a first addressable table that includes protected data associated with the subject and the encounter and at least a second addressable table that excludes protected data associated with the subject and the encounter. A set of addressable processes of the API are invoked to export one or more of: unstructured objects and structured data about objects from the first relational database, the structured data being completely absent of records associated with the subject and the encounter; unstructured objects and structured data about objects from the first and/or second relational databases, the structured data including records associated with the encounter but being completely absent of records associated with the subject; and unstructured objects and structured data about objects from the first and/or second relational databases, the structured data including records associated with the subject and the encounter.

[0010] Related method, systems and computer program products are also provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a block diagram illustrating a data ontology in accordance with some embodiments of the present inventive concept.

[0012] FIG. 2 is a nested block diagram illustrating the scope of observational data maintained within the data ontology in accordance with some embodiments of the present inventive concept.

[0013] FIG. 3 is a block diagram illustrating an application programming interface (API) having dual databases in accordance with some embodiments of the present inventive concept.

[0014] FIG. 4 is a diagram of a user interface and template for file mining in accordance with some embodiments of the present inventive concept.

[0015] FIG. 5 is a diagram illustrating the architecture and information flows of an API having multiple databases, front ends, storage locations, and authentication in accordance with some embodiments of the present inventive concept.

[0016] FIG. 6 is a block diagram illustrating an API having dual databases accessed by keys in accordance with some embodiments of the present inventive concept integrated in a storage unit of

a device.

[0017] FIG. 7 is a block diagram illustrating a data model, or Gene, for an exam in accordance with some embodiments of the present inventive concept.

[0018] FIG. 8 is a block diagram illustrating import functionality of an API in accordance with some embodiments of the present inventive concept.

[0019] FIG. 9 is a table listing DICOM tags used in accordance with some embodiments of the present inventive concept.

[0020] FIG. 10 is a block diagram illustrating an API including a method to publish, or export, a digital binder that contains a multiplicity of elements in accordance with some embodiments of the present inventive concept.

[0021] FIG. 11 is a block diagram illustrating a system in accordance with some embodiments of the present inventive concept.

[0022] FIG. 12 is a block diagram of a high level data processor that can be used in accordance with some embodiments of the present inventive concept.

[0023] FIG. 13 is a diagram illustrating operations of the A-side database in accordance with some embodiments of the present inventive concept.

[0024] FIG. 14 is a diagram illustrating operations of non-PII acquisition data stored in the B- side database schema in accordance with some embodiments of the present inventive concept.

[0025] FIG. 15 is a diagram illustrating a data import process performed by the API in accordance with some embodiments of the present inventive concept.

DETAILED DESCRIPTION OF EMBODIMENTS

[0026] The inventive concept now will be described more fully hereinafter with reference to the accompanying drawings, in which illustrative embodiments of the inventive concept are shown. This inventive concept may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the inventive concept to those skilled in the art. Like numbers refer to like elements throughout. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. [0027] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the inventive concept. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0028] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this inventive concept belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and this specification and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0029] As will be appreciated by one of skill in the art, the inventive concept may be embodied as a method, data processing system, or computer program product. Accordingly, the present inventive concept may take the form of an entirely hardware embodiment or an embodiment combining software and hardware aspects all generally referred to herein as a “circuit” or “module.”

Furthermore, the present inventive concept may take the form of a computer program product on a computer-usable storage medium having computer-usable program code embodied in the medium. Any suitable computer readable medium may be utilized including hard disks, CD-ROMs, optical storage devices, a transmission media such as those supporting the Internet or an intranet, or magnetic storage devices.

[0030] Computer program code for carrying out operations of the present inventive concept may be written in an object-oriented programming language such as Java®, Smalltalk or C++. However, the computer program code for carrying out operations of the present inventive concept may also be written in conventional procedural programming languages, such as the “C” programming language or in a visually oriented programming environment, such as VisualBasic or JavaFx.

[0031] The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer. In the latter scenario, the remote computer may be connected to the user's computer through a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

[0032] The inventive concept is described in part below with reference to a flowchart illustration and/or block diagrams of methods, systems and computer program products according to embodiments of the inventive concept. It will be understood that each block of the illustrations, and combinations of blocks, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the block or blocks.

[0033] These computer program instructions may also be stored in a computer-readable memory that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture including instruction means which implement the function/act specified in the block or blocks.

[0034] The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide steps for implementing the functions/acts specified in the block or blocks.

[0035] As discussed in the background of the present inventive concept, the management of medical images and data in the context of research and clinical trials is notoriously difficult even as the opportunities and demand for imaging biomarkers and artificial intelligence clinical decision support systems is rapidly expanding. Meeting the demands of image-driven innovation and clinical care in the era of big data and artificial intelligence (AI) generally requires a comprehensive ontology that covers the medical imaging domain from hardware definition to observation records, from subject to image, from anatomy to disease. The ontology must be supported by methods to store records and images, move images and data from devices to storage to application, methods to rapidly curate, visualize, and annotate images, trace the provenance of images and data through algorithm development and validation, maximize the sharing of images and data over space and time, and protect the rights of access and use of individual patient data as required both ethically and legally. Accordingly, some embodiments of the present inventive concept provide an application programming Interface (API) and directory management system for organizing and managing data records in a database and objects in a “data lake” that provides ontology needed for big data and AI as will be discussed further herein.

[0036] As used herein, AI refers to the simulation of human intelligence processes by machines, especially computer systems. Specific applications of AI include expert systems, natural language processing, speech recognition and machine vision. Some embodiments of the present inventive concept include a machine learning/AI component. The machine learning component can implement machine learning algorithms or artificial intelligence (AI) to generate and/or update

neural networks that are executed by a processor (FIG. 12). In some embodiments, the machine learning component can use one or more machine learning algorithms to generate one or more models or parameter functions for the detections. The machine learning component can be configured to generate an event model that understands which types of data indicate which types of detections. One or more of these event models may be used to determine an expected value or occurrence based on analysis of received data. In some embodiments, criteria can be designated by a user, admin, or automatically. By designating specific types of detections, resources (e.g., processing power, bandwidth, etc.) can be preserved for only the types of detections desired.

[0037] Various types of algorithms may be used by the machine learning component to generate the output desired. For example, certain embodiments herein may use a logistical regression model, decision trees, random forests, convolutional neural networks, deep networks, or others. However, other event detection models are possible, such as a linear regression model, a discrete choice model, or a generalized linear model.

[0038] Some non-limiting examples of machine learning algorithms that can be used herein can include supervised and non-supervised machine learning algorithms, including regression algorithms (such as, for example, Ordinary Least Squares Regression), instance-based algorithms (such as, for example, Learning Vector Quantization), decision tree algorithms (such as, for example, classification and regression trees), Bayesian algorithms (such as, for example, Naive Bayes), clustering algorithms (such as, for example, k-means clustering), association rule learning algorithms (such as, for example, A priori algorithms), artificial neural network algorithms (such as, for example, Perceptron), deep learning algorithms (such as, for example, Deep Boltzmann Machine), dimensionality reduction algorithms (such as, for example, Principal Component Analysis), ensemble algorithms (such as, for example, Stacked Generalization), and/or other machine learning algorithms.

[0039] These machine learning algorithms may include any type of machine learning algorithm including hierarchical clustering algorithms and cluster analysis algorithms, such as a k-means algorithm. In some cases, the performing of the machine learning algorithms may include the use of an artificial neural network. By using machine-learning techniques, copious amounts (such as terabytes or petabytes) of received data may be analyzed to generate models without manual analysis or review by one or more people.

[0040] An API refers to a way for two or more computer programs to communicate with each other. It is a type of software interface, offering a service to other pieces of software. A document or standard that describes how to build or use such a connection or interface is called an API specification. A data lake refers to a centralized repository that allows you to store all structured and unstructured data at any scale.

[0041] As further used herein, ontology refers to a set of concepts and categories in a subject area or domain that shows their properties and the relations between them. In the present inventive concept, the ontology refers to a hierarchical organization of entities that capture the data associated with a medical research program, clinical trial, or clinical evaluation, as illustrated in, for example, FIG. 1. At the top level of the hierarchy, data entities are organized by a set of entities that define the circumstances under which data is managed, a set of entities that define the data itself.

[0042] In this context, entities that define data management include entities that capture the creation of source data, for example through an examination of a patient or other subject, the movement of source data, and the access of source data, as will be defined further below.

[0043] In this context, entities that define the management of data includes entities that define Participant **100**, Study **140**, Location **150** and Encounter **170**. A Participant may be the subject of a source data generating procedure or may be associated directly or indirectly with managing the generation of source data from a subject. In our ontology, Subject **110** is divided into four categories: human subject **115**, animal subject **120**, a specimen **125** derived from an animate

subject, and inanimate subject **130**. The human entity is paired with an entity defining relations **117**. The animal entity is paired with a colony **122** and the colony may be divided into cohorts **124**. Specimens are paired with an Association **127** to parent subject. Inanimate object may be associated with a Lot **132** and the lot may be divided into batches **134**.

[0044] The group defining participants that are not the subjects are identified broadly as Examiner **135**, and the Examiner entity is again separated into Personnel **137** and Provider **139**. Personnel can be any person involved in any role associated with managing a procedure, and the roles are defined by a Type table associated with Personnel. The Provider entity provides the record of responsible individuals who receive, review, consult, or otherwise hold a position of accountability for the subjects and their procedures. A Provider may, for example, be a clinician or a principal investigator, and the type of Provider may be further specified in a Type table.

[0045] The Study **140** entity contains the definition of a set of examinations or investigations, which may include one or more subjects and one or more procedures that generate source data. The Study is divided into a set of Protocols **145**, and Protocols are further divided into a set of Projects **147**. The heart of the definition resides in the Protocol entity. The Protocol entity defines the set of required or allowed procedures, the inclusion and exclusion criteria of subjects, the roles and permissions of personnel associated with examinations or investigations. Projects are further segmentations of protocols that support the separation of responsibilities and workflows and provides finer granularity in the management of source-data provenance.

[0046] Workflows are managed through the Encounter **170** entity. The Encounter entity is further divided to manage multiple workflows important to management of Participants and Studies. The first workflow entity is the Visit **171** entity. The Visit provides for a set of Procedures **172** that are scheduled or performed at a specific point of time or stage of a Project (or Protocol or Study). Procedures are further sub-divided into Interventions **173**, Exams **174**, and Data Capture **175**. Interventions refer to a class of actions undertaken with respect to a Subject prior to the Exam. The Exam is the activity that generates results that are the target of the Procedure. Data Capture is the specific activity for preparing, securing, transporting, or storing the records associated with the Exam. A separate workflow entity is the Consent **176** entity. The Consent entity manages the set of documents, approvals, permissions, or constraints that may be required by the Project (Protocol or Study) for the conduct of the Visit, as well as access rights to the source data generated during a Visit. The Consent entity is vital to management of Personally Identified Information (PII), Protected Health Information (PHI), and maintaining compliance to Health Insurance Portability and Accountability Act (HIPAA) in the United States and general data protection regulation (GDPR) in the European Union.

[0047] The Encounter entity also includes the following workflow management entities: Communications (not shown), Third Party **177**, and Data **180** workflow entities. The communications entity provides structures for managing general communications, such as email, phone calls, facsimile, and the like. The Third Party **177** entity provides the structure for managing interactions with a) Collaborators **179**, and b) External parties **178**. As the name suggest, the Collaborator entity provides a structure for managing encounters with other parties who support an investigation or study that are outside of the primary group of Participants. External Party **178** encounters provide structure for managing encounters with, for example, auditors or regulatory bodies who in some way interact with the investigation or its source data.

[0048] The Encounter entity **170** provides a specific construction for managing interactions with source Data, including data derived from a Procedure and recorded at the time of Data Capture as well as other sources of data as described below. The data model further provides traceability of interactions with the Data through the Data Interaction **182** construct. The Data Interaction construct is divided to organize interactions with structured Data Records **183**, such as protected or unprotected Subject metadata, and the unstructured Data Objects **184**, such as image files. The data encounter log includes access, modifications, copies and transfers of source data, organized by

records (fields that can be written into a database or file) and objects (images and documents and the like). The data encounter also manages inflows and outflows from external electronic medical records systems **186**, included Picture Archiving and Communications System (PACS) **187** and electronic health records (HER) **188** Systems, as well as inflows and outflows from other Electronic Data Capture (EDC) **189** systems that may be feeders or receivers of data. Additionally, the data encounter provides an entity for Publications **185**, in order to trace the provenance of data into documentation.

[0049] A Location **150** entity is used to manage the organization and geographic location of activity. A parallel construct is maintained to separate organization for geography. For example, as shown in FIG. **1**, we may have multiple types of Institutions with unique hierarchies. A government agency may be defined by Department **155**, Agency **156**, Institute **157**, Branch **158**, and Sector **159**. An academic institution may be defined by University **160**, College **161**, Department **162**, Lab **163**, and Station **164**. A physical location may include an Address **165**, Campus **166**, Building **167**, Floor **168**, and Room **169**.

[0050] The aforementioned entities of the ontology describe the who (Participant), what and when (Encounter), where (Location), and why and how (Study). The observational output in this ontology is managed through a generalized Panel **190**. A Panel describes a set of observational records and objects derived from an exam that is part of Procedure undertaken for a Subject (by an Examiner). The Procedure **195** occurs during an Encounter **192** (for example, Visit), as part of a Project, defined by a Protocol that is part of a Study. For any given Procedure, a Form **196** is maintained to capture observational records **197**, and a Binder **198** is maintained to capture source objects **199**. The intent of the present inventive concept is that Forms and Binders are managed electronically, however they may be managed physically, copied and scanned into the data structure, or referenced by an electronic address to a physical location. The provenance of source data from its origin is maintained through the Participant, Study, Location and Encounter entities, where the Encounter Entities trace the forward flow of source data through Data Encounter entity (so long as the record is maintained within the platform).

[0051] The scope of a Panel is a feature of the present inventive concept. A Panel may start as a template, and the template is fully implemented in the context of a project fulfilling the requirements of a protocol as part of a study. As a record, a Panel will generally be specific to a subject (though one may envision a circumstance where a panel is recorded for a group), may span multiple encounters, with one or more procedures, one or more forms, and one or more binders, as illustrated in, for example, FIG. **2**.

[0052] As illustrated in FIG. **2**, a Panel **200** is a set of one or more observations derived from a Procedure **220** during an Encounter **210**. Each Procedure **220** may have one or more structured observations, or records **225**, that may be recorded on a Form **224**, where the Form **224** may be an electronic record in a database. Each Procedure may also result in unstructured objects **227** such as images, which are abstracted to reside in binders **226**, which may be electronic storage, PACS systems, Vendor Neutral Archives, and the like.

[0053] Panels are reusable and nestable constructs that define a set of target observations from reproducible examinations. Panels may be constructed at the lowest level of a single procedure with a single observational record or may be more complex constructs. Panels may likewise be built up of other panels. Thus, Panels form traceable data constructs within the data ontology of the present inventive concept, or Genes of the Data Genome (as discussed below). As further noted in FIG. **2**, Panels with observations are unique to Subjects **230** (though a Group may be a Subject if so defined).

[0054] A further hierarchy is introduced to manage the inherent complexity of biomedical research. A Study **260** is a top-level construct that may guide a multi-year, multi-site endeavor, for example, a Study may focus on the genetic origins of Glaucoma. In living subjects research, specific experimental forays into the Study are defined by Protocols **250** that are generally reviewed and

approved by Institutional Review Boards (IRB) or the like. A single protocol may allow for extended investigative branches, or Projects **240**. The encounters then of Subjects **230** are thus in the context of a Project **240** defined within a Protocol **250** inside of a larger Study **260**. This construct provides significant flexibility within a controlled system required for living subjects research. Other hierarchies may be applied without deviating from the intent of the present inventive concept.

[0055] It will be understood that the details illustrated in and discussed above with respect to FIGS. **1** and **2** are provided for example only and embodiments of the present inventive concept are not limited thereto.

[0056] The unstructured, semi-structured, and structured data that makes up the medical imaging and health record environment is very important to clinical and research development. As one of skill in the art understands, being able to access this data and manipulate it is very important for both the present treatment of patients as well as for developing new treatments for the future. To further complicate use of this data, patients are entitled to anonymity and thus all personal data must be stripped from the remaining data before this data can be shared, searched, manipulated etc. Accordingly, some embodiments of the present inventive concept provide an API that has a two-sided database. It will be understood that more than two sides may be implemented without departing from the scope of the present inventive concept. A first side of the database (A-side or Record-side) includes metadata describing the Subject, Encounter, Study and the like, as well as metadata describing the Exam and the Data captured during the Exam. The second side of the database (B-side or object-side data lake) includes information describing Data, for example image objects and all metadata associated with the image, but excludes identifying data/information associated with the Subject. Thus, the data on the B-side may be manipulated, searched, shared etc. without any concern for confidentiality, except the extent that an object itself is considered to be itself identifying of a subject.

[0057] Furthermore, the two sides of the database communicate through the API using A- side and B-side keys. This allows data from the B-side to be reassembled into the patient specific data if and when it is necessary. Thus, some embodiments of the present inventive concept address many of the disadvantages discussed in the background related to manipulating, searches, sharing etc. big data as will be discussed further below.

[0058] Referring now to FIG. **3**, a block diagram illustrating an API in accordance with some embodiments of the present inventive concept will be discussed. As illustrated in FIG. **3**, the API **300** operates between the Record-side Database **305** (A-side) and the object-side data lake **310** (B-side) as discussed above. In some embodiments the API **300** may be included with an imaging system itself (resident on the imaging device), however, it may also sit between devices without departing from the scope of the present inventive concept. The executable **315** portion of the API **300** may include its own user interface **320** and its own directory manager **325** including rules and permissions, methods (processes), log etc. Once the image is obtained and is received by the API, the image is stored by reference in the Database **330** (A-side) and the image itself is stored as an object in the Object-Side Data Lake **310**.

[0059] All metadata associated with the acquisition of the image and the image itself is stored according to a data model appropriate to the use case. This data model constructed in the context of an ontology **335** for a specific Exam that may include all or a subset of the ontology described in FIG. **1**. Herein, the broad data model for the acquisition and management of images and data in the context of medical applications is referred to as the Translational Imaging Innovations (TII) Data Genome. Each unique body of information forms a subset, or Gene, of the TII Data Genome. For example, each specific imaging system has a unique set of defining characteristics, and these characteristics are the Gene for the imaging. The ontology defines the scope of the data that may be managed and stored in the database and defines the scope of operations that are programmable from the API. The Genes provide low-level specific data models that may be “discovered” or added

to the data model and database over time without perturbing the structure of the data model.

[0060] Providing the database **330** in the Record-Side Database **305** removes the need for a developer to create a database of his/her own for each individual system. As further illustrated in FIG. **3**, the object-side data lake **310** includes the anonymized object storage records including objects **345** and object metadata **350**. The A-side and B-side databases can only access one another using unique keys associated therewith. Anonymization of data on the B-side simplifies development of a hardware system and essentially creates a “plug in” for the imaging device. A “plug in” generally refers to a software component that adds a specific feature to an existing computer program by essentially plugging in.

[0061] Users can create custom queries for the data stores using, for example, the user interface **320**. The user interface **320** can take any form known to those of skill in the art, for example, a keyboard, mouse, touch screen, voice command etc. The query structure in some embodiments may resemble a shopping list/cart. In other words, a user can choose from a list of items that identify what the user is looking for. For example, the list may include subject-what was imaged; what conditions was the subject imaged under; what records are available and the like. Each category has subcategories that can be checked and unchecked as the queries are refined.

[0062] Furthermore, the data may be stored in such a way that allows a user to reorganize the hierarchy of the queried data. In other words, a user can look at a folder structure with any hierarchy desired. Modules may be, for example, dragged and dropped or clicked, to indicate order. An example folder tree builder is diagrammed in the Export Collection **400** window of FIG. **4**. A list of Available Identifiers **401** defines available folders for data organization. The Folder Tree Builder **402** is a visual hierarchy of the structure given the ordering of the folders **401**. A user may add or delete levels according to the identifiers available in the user interface (UI).

[0063] A common hierarchy for a multi-study data record may include, for example, Study, Subject, Modality (e.g. imaging modality) and Date Range. A four level folder hierarchy might be as follows: Study/Subject/Date Range/Modality/{file objects}, but there are 24 ways to order this set of four attributes. Furthermore, it may be organized with only three levels or two levels depending on the user's needs. The construct of the current inventive concept allows the user to visualize data organization in folder structures that they are comfortable with, or that most closely resemble the way they are approaching the data for a specific task. This presentation structure is only a user interface, and does not actually move data internally, and therefore increases the flexibility of data organization without the need for copying data into new organizational constructs, or over thinking the folder structure to begin with.

[0064] Some embodiments of the present inventive concept provide for custom naming using tags from structure to maximize efficiency. For example, files may be named in accordance with user preference while maintaining an original name in the code not visible to the user. Thus, files can be recreated when necessary and can be put back in the original format with the original data. In other words, names are applied as a “label” on top of each file, but the original file is always there and is not altered.

[0065] As shown in FIG. **4**, a user may invoke a file name builder **403** on a graphical user interface and draw components of a file name from attributes drawn from the subject metadata and attributes drawn from the object (file) metadata. The file naming method may include error- identification methods, such as identifying names that are not unique or names that are too long for the operating system. A key advantage provided by embodiments of the present inventive concept is that informative file naming is allowed without transcription errors. Furthermore, as a file name is only a label, this construct allows for renaming files for specific use cases non- destructively. As shown in FIG. **4** (**403**), the file name is an ordered concatenation of encrypted subject ID, eye under test, wavelength of imaging system, image modality, and acquisition sequence number. This is quite useful for operating with image files where one wants to understand the contents at a glance. In other embodiments, the files may be delivered blind to data analysts who must remain unbiased. In

such a case, the files may be renamed with an arbitrary code or Universally Unique Identifier (UUID), while being fully traceable to the source through the database given appropriate authorizations.

[0066] Referring now to FIG. 5, a block diagram illustrating an architecture for the implementation of the API of FIG. 3 across multiple Record-side databases, object storage locations, and API front ends will be discussed. It will be understood that FIG. 5 is provided for example only and embodiments of the present inventive concept are not limited thereto. Embodiments illustrated in FIG. 5 illustrate how the API in accordance with embodiments herein may be able to manage a multi-client, multi-cloud (storage locations) environment with a common API and common object-side database (Referred to herein as OcuVault DB). The core of the API **500** controls methods that interact with structured records **501** and one or more databases **502**, reading and writing records to the main API database **503**.

[0067] The application layer **510** for the API interacts with the API through one or more endpoints (API endpoint App 1. . . API endpoint App_n). Between the application layer **510** and the API body **501** resides a validation layer **520**. Objects may be stored in any location **530** accessible to the API, including local to the API, in a generic cloud infrastructure such as Amazon Web Services, in a commercial storage structure such as Box or Dropbox and the like. File naming and folder organization problems are removed from the user and controlled by the API, using file naming and folder organization utilities, for example, discussed above, to facilitate visualization and access to data.

[0068] Storage Locations **530** in FIG. 5 of the present inventive concept suggest remote, networked, or cloud data architectures. However, in some embodiments, the API may also be resident on a data acquisition device, automate the service of building a database of data acquisition sessions, and automate the secure storage of objects directly onto a storage unit in communication with the acquisition device.

[0069] Referring now to FIG. 6, a block diagram illustrating an API **600** implemented to manage records and objects specific to an imaging or Exam device having dual databases accessible by A-side **621** and B-side **623** keys in accordance with some embodiments of the present inventive concept will be discussed. As discussed above, the API **600** does not have to sit on the device, it can sit between devices as a bridge without departing from the scope of the present inventive concept. Furthermore, various parts of the system illustrated in FIG. 6 can be at different locations and still have a similar functionality without departing from the scope of the present inventive concept.

[0070] As illustrated in FIG. 6, the API (referred to herein as OcuVault-on-Device **601**) includes a storage unit **617**, a hardware controller **641** and as user interface display **651**. Although the hardware controller **641** is shown as including system configurations **643**, system calibrations **646** and a system controller **648** and the user interface display is shown as including a subject data input **658**, an acquisition parameter input **656** and an acquisition control **652**, embodiments of the present inventive concept are not limited thereto. More modules may be added and/or removed without departing from the scope of the present inventive concept.

[0071] In the configuration shown in FIG. 6, ocuVault-on-Device **601** mediates a data management contract between an acquisition system and data storage. The contract may specify what data from the acquisition system **603** is to be temporarily stored (temporary storage **608**) and where, signal the API that an acquisition sequence is complete, and the API may then perform the service of collecting the data, filling the record-side and object-side databases (**603** and **606**), and stores the data objects, e.g., image objects **611** and image metadata **613**. In some embodiments of the present inventive concept, all PHI/PII shall stay resident in in the record-side database (A-side **603**). The object side database (B-side **606**) shall include a reference to the storage path of the object and a set of defining attributes, agreed in the contract, for querying the object-side database (B-side **606**). The record-side database (A-side **603**) shall be linked to the object-side database **606** through an

exam (time-point-of-acquisition) key. Furthermore, all metadata associated with object necessary to read and analyze the object shall be recorded in a java script object notation (JSON) or similar record associated directly with the object. In some embodiments of the inventive concept, the object shall be stored in one of two formats: either a proprietary format explicitly defined by the customer, or a digital imaging and communications (DICOM)-conformant format. Metadata associated with object includes configuration and calibration information associated with the acquisition of the object. Metadata associated with the object includes all DICOM-conformant tags provided by the customer in the contract.

[0072] In some embodiments, ocuVault-on-Device **601** is installed as a Restful API on an examination device. A RESTful API refers to API or web API that conforms to the constraints of REST architectural style and allows for interaction with RESTful web services. A data model for examination is pre-defined as a subset of a master data model, and the data model has a mapping to a relational database managed by the API.

[0073] At the conclusion of an examination, or a single data acquisition of a patient or subject with the device, the device software writes any examination metadata and observational records to a tagged file that is consistent with the data model and saves the file to a predetermined location. The tagged file may be a JSON file, and XML file, a CSV file, or the like without departing from the scope of the present inventive concept.

[0074] The data files include any or all the structured records that are associated with the examination, available from the device, and consistent with the data model. These records may include patient metadata, examination metadata, and notes. These records may be available on the device because they have been, for example, downloaded through an external records system, such as an EHR system, read in via a barcode, or input through a graphical user interface (GUI).

[0075] The structured record may also include any or all of the device configuration settings that determine the device operation, computations, displays, or any other behaviors that support the record of the examination.

[0076] The structured record may also include information specific to accessories used during the preparation or conduct of the examination. Such accessories, without limitation, may be, for example, a specific objective lens used during imaging, a record of an intervention, such as pupil dilation, applied prior to or during the examination and the like.

[0077] The record may further include information on critical subsystems of the device. For an optical imaging system, subsystems may include, for example, light sources, detectors, spectrometers and the like. Information related to such subsystems may include, for example, serial numbers, calibration data, firmware versions that may provide information on the history of the subsystem that supports quantitation analysis that depends on the subsystem and/or related to the quality record of the subsystem.

[0078] The structured record is defined a priori for the device, and the system software only needs to populate data into a linear file with a known set of tags for each record at the conclusion of each acquisition.

[0079] Referring now to FIG. 7, an example data model or Gene that may be used for an exam in the context of the present inventive concept will be discussed. As illustrated, the data model includes the A-side **700** and B-side **770** construct discussed above with respect to the system. As further illustrated, the A-Side data **700** include structures for Subject **710**, Study **720**, and Personnel **730**. As used herein, Personnel refers to, for example, physicians and staff that conduct an examination. The A-side data also includes records pertaining to the Encounter **740**, the device or instrument used in an exam **750**, and the location of the exam **760**.

[0080] In the data model illustrated in FIG. 7, the B-side construct **770** includes information pertaining to the data acquisition, including parameters that define the acquisition **780**, parameters that define the state of the device at time of acquisition **790**, and specific device and subsystem settings at the time of acquisition **795**.

[0081] In addition, the device saves any objects associated with the exam, independently of, but along with the structured record file. Such objects may include, for example, formatted reports, spreadsheets, image files, configuration files, or other types of information without restriction. The file types may be, for example, portable document format (PDF) files, Microsoft Word files, tag image file format (TIFF) files, joint photographic experts group (JPEG) files, portable network (PNG) files, DICOM files, and the like without departing from the scope of the present inventive concept. Furthermore, the device may save source data, and the source data may be one or more data files that are derived from the source, and the source files may be raw files that are unprocessed, subjected to specific pre-processing steps, or may be processed to final state as determined by the device manufacturer for the intended use.

[0082] For example, a digital camera may output a color image into a digital negative (DNG) file or a JPG file. The DNG file is "source data" for subsequent processing and enhancement in a digital darkroom, such as Adobe Lightroom or Capture One. But the DNG file has already been processed from sensor pixel data using proprietary color science algorithms of the camera vendor. The color science algorithms provide the interpolation and smoothing required of a sensor that is covered by a specific color filter, such as a Bayer filter (U.S. Pat. No. 3,971,065) or a Fujifilm X-Trans sensor (U.S. Pat. No. 8,531,563) to produce a pleasing full color response.

[0083] In embodiments using color digital cameras, it may be desirable to save the pixel intensity values directly from the sensor prior to application of the color science algorithms, in addition to or instead of, the processed DNG (Adobe digital negative format).

[0084] The associated data model for a digital camera may include definition of the color filter array disposed over the sensor and may further include a calibration function that maps spectral efficiency function to specific pixel values. The data model may also include a distortion map associated with the sensor array alone or in combination with a lens used to acquire an image.

[0085] Imaging processing pipelines for monochrome or color digital sensors are discussed, for example, in U.S. Pat. No. 8,330,772 entitled IMAGE SIGNAL PROCESSOR FRONT-END IMAGE DATA PROCESSING SYSTEM AND METHOD. U.S. Pat. No. 8,743,229 entitled IMAGE PROCESSING METHOD AND APPARATUS FOR BAYER IMAGES describes a set of processing steps to produce an image from a sensor that includes a Bayer filter. U.S. Pat. No. 5,606,365 entitled INTERACTIVE CAMERA FOR NETWORK PROCESSING OF CAPTURED IMAGES discusses processing or correcting a color image with information downloadable from a database. Unique to the present inventive concept, the source data at the pixel level accompanies a set of metadata, calibration information, and error correction information in an electronic binder, or traveler that flows with the recorded object from the time of acquisition. Such low-level data may originate with the camera manufacturer or may have been updated through by the manufacturer, camera owner, or other third-party. This data need not be open sourced to every user. Encryption and password protection may be applied to limit access without deviating from the objective of the inventive concept to provide immediate access to authorized user of low-level data for maintaining provenance of data or enhancing post-processing and analysis of source data.

[0086] Another example of providing a precursor image can be found in the field of Fourier Domain Optical Coherence Tomography (FDOCT). The depth-dependent spatial cross-sectional image is formed from the Fourier Transformation of a wavelength-dependent interferogram, following a sequence of pre-processing steps, as discussed for example, in U.S. Pat. No. 8,401,257 entitled METHODS, SYSTEMS AND COMPUTER PROGRAM PRODUCTS FOR PROCESSING IMAGES GENERATED USING FOURIER DOMAIN OPTICAL COHERENCE TOMOGRAPHY (FDOCT). The clinically relevant image object is the spatial domain image. The process of Fourier transformation destroys phase information that is present in the original spectral domain, for example, wavelength dependent data.

[0087] It is often valuable to preserve the spectral domain data. To be useful, the spectral domain data requires certain calibration information that is used to complete the Fourier transformation.

This calibration information includes the mapping of the correct wavelengths to the spectral domain pixel values, and may further include dispersion compensation parameters that correct for the optical system of the device or the subject under test, or both.

[0088] In some embodiments of the present inventive concept, the spectral domain data is preserved as an object of the examination, and the spectral calibration and dispersion coefficients for the device and the subject under test are preserved in the structured metadata file.

[0089] In some embodiments of the present inventive concept only the spectral domain object and necessary calibration coefficients are stored; the Fourier-transformed spatial domain image is not stored immediately, and rather can be recreated at a later stage with spectral domain object and the associated metadata.

[0090] A particular value of preserving such image data in the spectral domain is in securing PHI from unauthorized access. FDOCT is commonly used in retinal imaging, and an image of the retina is PII and PHI. The spectral domain precursor is not PII or PHI, and requires processing with the associated metadata, or calibration and compensation factors to recover an image. This metadata may be protected through encryption in the metadata file, such that only authorized recipients can complete the processing steps to generate the spatial domain image. While this method does not offer foolproof security, it will be understood that sufficient a priori knowledge exists to constrain a search for a set of processing parameters that recovers the target spatial domain image, it does prevent incidental disclosure of PHI, while preserving information that may be useful in recovering phase information, phase variation over time that may accompany blood flow, and in assessing spectroscopic information associated with the patient or subject under test.

[0091] Referring now to FIG. 8, a flow diagram illustrating the import functionality of the API in accordance with embodiments of the present inventive concept will be discussed. As illustrated, after an acquisition **810** is completed, the API import **801** is initiated, a set of structured records is written to a tagged file (e.g. JSON) and a set of file objects is saved with the JSON file a target location **810**. The API validates **820** the JSON record and then validates the files referenced by the JSON record. Errors, if any, are written to a log file **821**.

[0092] The structured file is read by the computer code, and the tagged records are recorded to a relational or hybrid database accessible to the API **830**, where the database schema is precast as the blueprint, or genome for the data model, including not only the sets of tables and fields to hold the data, but includes the common relational structures as well for recording relationships of studies, patients, encounters, and examinations that form the context of the broader model.

[0093] The API code stores the tagged file **840** or a copy thereof into a secure storage location, referred to as the A-Side storage discussed above. As discussed, A-side storage may retain PHI as a source record.

[0094] The API creates a second copy of the tagged file, stripped of all PHI, and stores this copy **850** to B-Side storage as discussed above. The tags themselves, by design of the API and data model, define PHI vs non-PHI information, and therefore no additional algorithm intelligence is required to parse the tagged file into the A-side and B-side counterparts.

[0095] The API stores the path to the object files, either according to their original location or after a copy or move action according to system preferences, in the database.

[0096] The API performs a validation that the operations were successfully completed. This validation may include, for example, confirmation that the tagged fields are appropriately formatted, and that object files are in their correct location.

[0097] The API sends a message **860** to the device that the operations were completed, provides a log of actions undertaken, and a log of error messages for the case that validation failed **821** (Return 400 Error). The API returns to its RESTful state pending the next examination or acquisition.

[0098] In some embodiments of the present inventive concept, the system software of the device may invoke the API, the API may read for data in the target storage location, or the API may be

invoked periodically using a scheduler. Other triggers for invoking the API may be appropriate to a particular use case, without deviating from the intent of the inventive concept.

[0099] In addition to these data ingestion methods of the API, the API may include other methods useful to the transfer and utilization of the records, objects, and database managed by the API. For example, the API may automatically create a structured file object by combining structured records stored in the tagged file with one or more of the unstructured file or image objects. A digital camera image file may be converted to a DNG file by combining the image object with tagged metadata into an exchangeable image file (EXIF)-equivalent data structure. A medical image may file may be converted into a DICOM file by converting the source object into the DICOM image format and mapping the tagged metadata record to DICOM tags.

[0100] A distinct advantage provided by the present inventive concept over the conventional systems and methods is the extensibility of the conversion methods to source objects not originally envisioned by conventional approaches. For example, in accordance with embodiments of the present inventive concept, a DNG file may be created by invoking the API to create a two-dimensional (2D) cross section from a volumetric Optical Coherence Tomography (OCT) image and publishing this extracted 2D image to a DNG-conformant file.

[0101] Another distinct advantage of the present inventive concept over the conventional systems and methods is the automation of publishing source data to DICOM conformant files, even where a specific DICOM standard or supplement does not yet exist. In such a circumstance, any data that may be rendered as one or more frames of an image may be converted into DICOM-readable format, either as direct representation of the source data or after a post-processing step. The data model of the present inventive concept is expressly mapped to DICOM tags as follows. DICOM reserves even-numbered tags as public tags and odd-numbered tags as private tags. Among the even numbered public tags, some are defined as mandatory (to claim DICOM conformance) and some are optional.

[0102] Operations of the API according to some embodiments of the present inventive concept include populating DICOM tags with the API metadata tags according to the following rules: [1] Populate mandatory public tags with available metadata; [2] Populate optional public tags with available metadata; and [3] Populate private tags using proprietary metadata drawn from the data model of the present inventive concept into a private, but common, set of DICOM tags reserved for the API (as allowed by the DICOM standard), as listed, for example, in the table of FIG. 9. It will be understood that the Table in FIG. 9 is provided as an example only and embodiments of the present inventive concept are not limited thereto.

[0103] A further advantage over conventional systems and methods is the automation of publication of a DICOM readable file independent of the source if whatever metadata that is made available is written into a tagged file according to the present inventive concept. The API handles public versus private information and handles PHI versus non-PHI information. A simple set of preferences guides the DICOM publication process, for example, “include (or exclude) PHI,” “include (or exclude) private information,” “Encrypt PHI,” and similar preferences.

[0104] Furthermore, a DICOM file is automatically tied to a relational database, and the relational database at its source is common to every other relational database in a common family built and according to the present inventive concept. Thus, while define and implement a data model for the management of medical, and specifically ophthalmic images, many other data models that are similar in detail or a level of abstraction may be envisioned and created that offer the benefits of the present data model and its associated API, without deviating from the intent of the inventive concept.

[0105] In some embodiments of the present inventive concept, the API may be used to publish a secure, traceable, and tamper-resistant data object for the transfer of examination data to an authorized receiver. In the field of clinical trials, for example, source document control is required. It is essential to ensure the provenance of data, trace data to its source, and ensure that such data

has not been modified on its journey.

[0106] In some embodiments of the present inventive concept, the API **1000** may publish, or export, a digital binder that contains a multiplicity of elements, as illustrated in, for example, the flow diagram of FIG. **10**. The elements may include a first tagged file of metadata associated with the examination. This tagged file may be the A-Side file discussed above, or a subset, and may include PHI, coded information tied to PHI, or may be free of PHI. The binder may include a second tagged file of metadata associated with the examination. This tagged file may be the B-Side file discussed above, or a subset, and will be free of PHI. The binder will additionally include any source objects that are useful to the processing and analysis of the examination record. These source objects may be the source image file and may additionally include configuration files or any other files generated during the examination that are captured and managed by the API. The record of each such stored object includes a universally unique identifier (UUID) stored in the database, and these are traceable objects.

[0107] Referring now to FIG. **10**, the export process by the user or system invoking the API to initiate an export **1010** with required set data IDs as may be identified from the device user interface or a query to the API database. The API validates **1020** that the IDs are valid, and that data is available, locating the subject data **1021** from the A-side database and the file object metadata **1022** from the B-side database.

[0108] Two JSON (or equivalent) files are generated from the database for the data to be exported: a file containing object metadata only **1031** and a file containing both object and subject metadata **1032** (the latter may exclude or code PHI).

[0109] A checksum **1041** is created for each such object using technologies known in the art, such as the SHA256 Hash Generator, and the resultant checksum is stored along with object in the binder, i.e. a zip file **1050** or similar. The hash is now identified with the UUID and may also be stored in the database or distributed separately to the authorized data recipient, and thus provides a traceable record for tamper-testing.

[0110] The binder with its objects, files, and hash codes may be compressed and may be encrypted into a single electronic binder, for example as a zip file, for transport to an authorized recipient.

[0111] Upon receipt, a counterpart to the API can perform automatic validation of the object. The hash codes can be tested to ensure that the independent files have not been corrupted. The tagged files can be read for critical information such as patient/subject under test, exam date, and the like. Any observational records can be read to compare to competing record that may have been transcribed separately into a different electronic data capture (EDC) system.

[0112] Further, the data binder contains all (most) information relevant to the source object, such that the binder may be imported automatically into the recipient's database, having been constructed according to the model of the present inventive concept and according to the common data model associated with this use case.

[0113] In some embodiments of the present inventive concept, the API may provide publication of PHI-free examination results for analysis, with all information required for analysis in one object or one such binder, drawn from public and private tags as necessary. This embodiment automates deidentification of data so necessary to the processing and analysis of human subject data, and necessary to any blinded analysis whether involving human subjects or not. UUIDs preserve traceability back to the source.

[0114] In some embodiments of the present inventive concept, the electronic binders are distributed to the recipient from the source over a network via one or more pre-defined servers, where the record of source, destination, and UUID of the binder (and/or UUID of each independent object in the binder with or without its hash code) is recorded at the server, as illustrated, for example, in FIG. **11**.

[0115] Referring to FIG. **11**, multiple sites **1100** contribute data via their local API export into secure binders **1110** that are transmitted over a network through a dedicated server **1120** that

records UUIDs of the binders and internal objects, the source IP address, and a timestamp. Additional information may also be recorded, including the binder size (e.g. in MB or GB), number of objects within the binder, and so on. This transport path provides immediate traceability, and as no information is read from the electronic binder other than UUID, preserves privacy and security. The binders are finally transported to a study coordinating center **1140** for import into a compatible master database, or catalog, **1130**.

[0116] As is clear from the embodiments discussed above, some aspects of the present inventive concept may be implemented by a data processing system. The data processing system may be included at any module of the system without departing from the scope of the preset inventive concept. Exemplary embodiments of a data processing system **1230** configured in accordance with embodiments of the present inventive concept will be discussed with respect to FIG. **12**. The data processing system **1230** may include a user interface **1244**, including, for example, input device(s) such as a keyboard or keypad, a display, a speaker and/or microphone, and a memory **1236** that communicate with a processor **1238**. The data processing system **1230** may further include I/O data port(s) **1246** that also communicates with the processor **1238**. The I/O data ports **1246** can be used to transfer information between the data processing system **1230** and another computer system or a network using, for example, an Internet Protocol (IP) connection. These components may be conventional components such as those used in many conventional data processing systems, which may be configured to operate as described herein.

[0117] Various embodiments of the present inventive concept have been discussed. These embodiments are meant as examples only and do not limit the overall inventive concept. Some embodiments of the present inventive concept provide an automated method to receive images and metadata from an acquisition device, parse metadata into data entities based on a standardized ontology, automatically write the data acquisition record into an acquisition side database, format images into a standardized image format, store images onto recording medium, include in the image format a set of information necessary to visualize and quantify the image, populate a storage record into a storage side database, link the records between the acquisition side database and the storage side database with a unique code.

[0118] As discussed, in some embodiments, the acquisition side database may include data elements associated with the subject under test and the condition of test, including any personally identifiable information (PII) of the subject under test, where PII includes protected health information (PHI), and where the storage side database excludes all PII.

[0119] Referring now to FIG. **13**, a diagram illustrating operations of the A-side database in accordance with some embodiments of the present inventive concept. As illustrated in FIG. **13**, the A-side database schema stores sensitive and PII-related data based on uniquely identified subject encounters **1301** with a one-to-many relationship to discrete subject visits **1302** at specified timepoints **1303**. When a procedure **1304** is performed during a visit **1302**, it is uniquely identified to a specific device instance **1305** by a universally unique identifier (UUID). The device instance **1305** data encapsulates not only information about the device itself, but also its physical location within an institution **1306**, the associated study **1307** or studies it is used, and the individual experiments **1308** conducted within each study. Each subject **1309** has data stored under 2 logically separated levels: coded data **1310**, and PII data **1311**. Therefore, when accessing the database, a service or user is either granted access to subject data through the coded table **1310** where all PII fields are obfuscated, or through the PII data table **1311** where all PII fields are fully accessible. Finally, the group of associated subject, procedure, and data sets for the encounter is uniquely identified and stored in the subject_metric table **1312**.

[0120] In some embodiments, all information necessary to analyze the image may be stored in a data record directly accessible with the image. The data entities may include the following: definition of the imaging method; definition of the hardware used to acquire the image; definition of configurations and setting used in the image acquisition; and calibration information associated

with quantitative interpretation of the image.

[0121] Referring now to FIG. **14**, a non-PII acquisition data stored in the B-side database schema is shown. Each acquisition **1401** is defined by its type, associated procedure, and complete configuration of the device used. Each acquisition is made up of channels **1402**, potentially including multiple types and modalities. A channel **1402** is associated one-to-many with captures **1403** of various types and parameters stored in JSONB (binary JSON format) data, which is associated many-to-many with stored file objects **1405** through the capture_file_object table **1404** allowing tracking of deeply-nested relationships among the current, and potentially future, file objects **1405**. For the acquisition, the complete state of the device and its configuration is uniquely identified, stored, and referenced by the device_configuration_state table **1406**, and its child-tables that store each device setting **1407**, subsystem configuration state **1408**, subsystem setting **1409**, component configuration state **1410**, and component setting **1411**. These individual device, subsystem, and component settings are then uniquely identified as a group in the capture_settings table **1412**, and associated with each capture within the acquisition.

[0122] In some embodiments, the ontology of the acquisition side database may include the following entities: a subject under test entity; an anatomy under test entity; a test method entity; an imaging system hardware entity; an imaging system software entity; a study entity; a time and date entity; and an examiner entity.

[0123] In some embodiments, the Subject Under Test Entity may be divided into two sub-entities, where the first sub-entity is absent of all PII, and where the second sub-entity includes PII.

[0124] In some embodiments, the PII sub-entity may be encrypted and accessible only with enabled permission.

[0125] In some embodiments, method may be performed on a system having an integrated directory management system.

[0126] In some embodiments, the directory management system may include a visual query interface for filtering entity attributes contained within the acquisition side database ontology and pairing with associated objects recorded in the storage side database.

[0127] In some embodiments, the results of the visual query interface are displayed in a multi-level hierarchical format, and where the order of the hierarchy may be set by the user in the visual query interface.

[0128] In some embodiments, the visual query interface may display a hierarchical structured view of mutually exclusive sets of selection attributes drawn from the data ontology, where the coarsest set of attributes for a comprehensively exhaustive set of attributes, and where within the visual query interface the user may indicate the hierarchical order for presenting the results.

[0129] In some embodiments, the set of attributes displayed below the coarsest level may be drawn from key-value pairs defined within the database ontology.

[0130] Some embodiments of the present inventive concept provide an automated method to link image and document objects with database records from a first object storage location housing image and document objects and a second storage location housing a database with records, though an Application Performance Interface (API), where the API automatically populates an object-side database, where the object-side database includes a coded-link between each object and a unique event linked to a subject in the record-side database.

[0131] The class structure illustrated in FIG. **15** illustrates the data import process performed by the API. When the ImportMain class **1501** is called, a structured JSON file is specified and the file data is parsed according to its relation to PII, where it's sent to the ImportASide class **1502**, or non-PII acquisition data, where it's sent to the ImportBSide class **1503**. Upon import **1501**, and after the data is parsed by the A and B side classes (**1502**, **1503**), each set of data, A-side and B-side, goes to the Validatable interface **1504** where the data is inspected for valid structure, completeness based on previously defined encounter parameters, known device parameters, dates, file paths, and so on. After validation, universally unique identifiers (UUIDs) are generated each data object within each

set of data, A-side and B-side, then standardized and stored internally by ocu Vault using the ExportMain class **1505**.

[0132] In some embodiments, automatically populating an object-side database may include searching a target data lake for objects that match the results of a query initiated from the record-side database, identifying an object in the target data lake, automatically populating a storage path identifier for the object into the object-side database, and returning to the record-side database a key that links a subject record in the record-side database to the object record in the object side database.

[0133] In some embodiments, populating an object side database may include populating the database with additional attributes descriptive of the object drawn from the object, including a method type that defines the creation of the object, a hardware definition that defines equipment used to generate the object, a configuration set that defines the operating conditions at use in the creation of the object, a calibration set that provides for quantitative interpretation of the object, as well as including one or more of a file type, a file size, a file creation date and time stamp.

[0134] In some embodiments, populating an record-side database may include populating the database with additional attributes descriptive of the object that are drawn from the record, including one or more of an activity type that defines the event creating the object, an anatomy or component of the subject that is the subject of the object, a gender of the subject, an age of the subject at the time of the object curation, and a pathological diagnosis or condition of the subject at the time of object creation, a method type that defines the creation of the object.

[0135] In some embodiments, the process of linking the object to the database record may be preceded by one or more actions invoked by the API to modify an object.

[0136] In some embodiments, the modification of the object may include encrypting or removing protected health information from the object.

[0137] In some embodiments, the modification of the object may include reformatting the object into a new format.

[0138] In some embodiments, reformatting may include removing header information from the existing format and combining the header information with information from the record-side database into a new metadata file that is itself stored locally and linked to the object.

[0139] In some embodiments, modifying the object may include making a copy of the object, and performing the modification only on the copy of the object, leaving the original copy of the object in its original state.

[0140] In some embodiments, the API may include an integrated directory management system.

[0141] In some embodiments, the directory management system may include a visual query interface for filtering entity attributes contained within the record-side database ontology and pairing with associated objects recorded in the object side database.

[0142] In some embodiments, the results of the visual query interface may be displayed in a multi-level hierarchical format, and the order of the hierarchy may be set by the user in the visual query interface.

[0143] In some embodiments, the visual query interface may display a hierarchical structured view of mutually exclusive sets of selection attributes drawn from the data ontology, where the coarsest set of attributes for a comprehensively exhaustive set of attributes that define the provenance of source data and the structure of observational records and objects that form the source data, and where within the visual query interface the user may indicate the hierarchical order for presenting the results.

[0144] In some embodiments, the set of attributes displayed below the coarsest level may be drawn from key-value pairs defined within the database ontology.

[0145] Some embodiments of the present inventive concept provide an Application Programming Interface for managing the connection if image objects to database records, where the API includes methods and a user interface, where at least one method automatically constructs a database of

metadata representative of the image objects and the user interface provides a graphical display of the contents of the image objects referenced in the database.

[0146] In some embodiments, the method for constructing the database may include creating and storing a coded name of the image object, creating a file object with additional metadata package relevant to interpretation of the image object, storing the file object in proximity to the image object, and providing a union of the image object and its metadata package with a mutual reference to the path of this object union, storing a path in the database to this object union.

[0147] In some embodiments, the method for constructing the database may include providing a coded link to a subject database to a subject, the subject database containing metadata relevant to the subject of the image object and the creation of the image object.

[0148] In some embodiments, the API user interface comprises an integrated directory management system.

[0149] In some embodiments, the directory management system includes a visual query interface for filtering entity attributes contained within the subject-side database ontology and pairing with associated objects recorded in the object side database.

[0150] In some embodiments, the results of the visual query interface are displayed in a multi-level hierarchical format, and the order of the hierarchy may be set by the user in the visual query interface.

[0151] In some embodiments, the visual query interface may display a hierarchical structured view of mutually exclusive sets of selection attributes drawn from the data ontology, where the coarsest set of attributes for a comprehensively exhaustive set of attributes, and where within the visual query interface the user may indicate the hierarchical order for presenting the results.

[0152] In some embodiments, the set of attributes displayed below the coarsest level may be drawn from key-value pairs defined within the database ontology.

[0153] Some embodiments of the present inventive concept provide a directory management system for visualizing and managing objects in a database or data lake according to a data ontology.

[0154] In some embodiments, the data ontology may be defined by a mutually exclusive and comprehensively exhaustive set of entities appropriate to a domain of data to be managed.

[0155] In some embodiments, the domain may be medical informatics, and the entities may include subjects, diagnosis codes, examiners, examination dates, and examination modalities.

[0156] In some embodiments, the domain may be medical informatics, and the entities may be subjects, research studies, diagnostic hypotheses, investigators, investigation dates, and investigation modalities.

[0157] In some embodiments, domain may be medical informatics, and the entities may include examination modalities, where examination modalities further include imaging or functional test modalities, where the imaging or functional test modalities contain further reference to the provenance of the test instruments, where the provenance to the test instruments includes the manufacturer and model of the test instrument.

[0158] In some embodiments, the provenance to the test instrument further comprises at least one of a set of instrument configurations and a set of instrument calibrations records appropriate to an examination.

[0159] In some embodiments, the entities may integrate a library of multiple disparate entities drawn from external sources, and where the external-drawn libraries include entities that standardize data definitions.

[0160] In some embodiments, the libraries may include one or more of a library of devices registered by a regulatory body and a library of common data elements defined by a domain-relevant data standards body. Note: this may include reference to FDA GUDID (Global Unique Device Identifier), GMDN (Global Medical Device Nomenclature), LOINC (Logical Observation Identifier Names and Codes), SNOMED, NIH CDE (Common Data Elements).

[0161] In some embodiments, the directory management system may include a visual organization structure, where the visible organization structure includes blocks of information directly drawn from and referenced to selected addressable entities of the data ontology.

[0162] In some embodiments, the addressable entities of the ontology may be selected programmatically as a subset of available entities from the ontology, which may be a complete set, through an Applications Programming Interface, and where the API may expose the full set or partial set of addressable entities through an executable program, a command line interface, or a graphical user interface.

[0163] In some embodiments, the addressable entities may be presented on a graphical user interface or accessible programmatically for down-selection, where down-selection comprises selecting a subset of addressable entities for organizing a collection of data, where the subset may be one, some, or all, of the addressable entities.

[0164] In some embodiments, the collection of down-selected addressable entities may be presented to the user on a display, where the user can multi-select one or more of the presented entities, where the selected entities are presented as individual blocks or panels on a user interface, where each individual panel includes one or more searchable attributes of the entity, where the user can multi-select one or more values of an attribute in each individual panel, and where the intersection of selected attributes of entities forms a complete query on the data in the database or data lake, and where a result of the query is made available to the user.

[0165] In some embodiments, a presentation of entity attributes may be constrained by a set of user permissions to allowable attributes, where allowable attributes are further constrained to be show actual values or coded values, such that a user may be blocked from seeing identifiable values that are forbidden according to the rules of the permission set.

[0166] In some embodiments, the values of data records or names of objects that are returned by the query may be constrained by the same set of user permissions, such that a user may be blocked from seeing identifiable values in data records or object names that are forbidden according to the rules of the permission set.

[0167] In some embodiments, the data contained within objects that are returned by the query may be constrained by the same set of user permissions, such that data contained with objects is redacted or coded, such that a user may be blocked from seeing identifiable values in with objects that are forbidden according to the rules of the permission set.

[0168] In some embodiments, the user may invoke one or more methods to operate on the results of the query.

[0169] In some embodiments, the method may include the renaming of objects returned by the query, where such renaming is defined by a template that includes components drawn from a selected attribute of the selected addressable entities.

[0170] In some embodiments, the template may be modified by the user within an API, where the query and the renaming template may be stored and recalled on demand or on a schedule, and where recall includes the updating of the results of the query based on changes to the database or data lake.

[0171] In some embodiments, the flexible directory management system may include the results of query on a subset of addressable entities and a subset of addressable values of such entities, where the set of addressable entities is defined within the API by reference to a data ontology, where the visibility of attributes may be constrained through coding or redaction according to a set of rules that set user permissions, where the names of objects returned by the query may be modified on display according to a template drawn from the ontology, addressable entities, and addressable values, where the naming convention is subject to the constraining set of rules.

[0172] In some embodiments, the results of the query may be organized in an exploded or indented hierarchy, where the levels of the hierarchy correspond to the selected addressable entities of the data ontology.

[0173] In some embodiments, the order of the hierarchy may be modified on demand in any of the $n*(n-1)$ ordered combinations of any n selected addressable entities that form the basis of the query.

[0174] In some embodiments, the hierarchy may be displayed as a nested set of folders, where the folders are named according to the differentiated attribute of an entity, and where the objects returned in the query are in the lowest folder of the hierarchy, where if the lowest folder level in the hierarchy has a trivial contribution to the query, the objects returned in the query will reside in the lowest folder level with a non-trivial contribution to the query. As used herein, “trivial” refers to no further differentiation of data afforded by the query result defining the folder level.

[0175] In some embodiments, the directory management system may be set to display one or more views of resident objects according to the results of a specified query, in a hierarchical structure that corresponds to a subset of addressable entities within the ontology of the database or data lake, where the various views are available to the user according to the various demands of the user, subject to any rules and permissions that are established for viewing of the data and data objects.

[0176] In some embodiments, each hierarchical view of resident objects may provide a comprehensively exhaustive view of the available resident objects, subject to constraints of the rules and permissions.

[0177] In some embodiments, the methods may be non-destructive of the data and objects within the database or data lake, and where the history of invoked methods is recorded in an audit log.

[0178] The aforementioned flow logic and/or methods show the functionality and operation of various services and applications described herein. If embodied in software, each block may represent a module, segment, or portion of code that includes program instructions to implement the specified logical function(s). The program instructions may be embodied in the form of source code that includes human-readable statements written in a programming language or machine code that includes numerical instructions recognizable by a suitable execution system such as a processor in a computer system or other system. The machine code may be converted from the source code, etc. Other suitable types of code include compiled code, interpreted code, executable code, static code, dynamic code, object-oriented code, visual code, and the like. The examples are not limited in this context.

[0179] If embodied in hardware, each block may represent a circuit or a number of interconnected circuits to implement the specified logical function(s). A circuit can include any of various commercially available processors, including without limitation an AMD® Athlon®, Duron® and Opteron® processors; ARM® application, embedded and secure processors; IBM® and Motorola® DragonBall® and PowerPC® processors; IBM and Sony® Cell processors; Qualcomm® Snapdragon®; Intel® Celeron®, Core (2) Duo®, Core i3, Core i5, Core i7, Itanium®, Pentium®, Xeon®, Atom® and XScale® processors; and similar processors. Other types of multi-core processors and other multi-processor architectures may also be employed as part of the circuitry. According to some examples, circuitry may also include an application specific integrated circuit (ASIC) or a field programmable gate array (FPGA), and modules may be implemented as hardware elements of the ASIC or the FPGA. Further, embodiments may be provided in the form of a chip, chipset or package.

[0180] Although the aforementioned flow logic and/or methods each show a specific order of execution, it is understood that the order of execution may differ from that which is depicted. Also, operations shown in succession in the flowcharts may be able to be executed concurrently or with partial concurrence. Further, in some embodiments, one or more of the operations may be skipped or omitted. In addition, any number of counters, state variables, warning semaphores, or messages might be added to the logical flows or methods described herein, for purposes of enhanced utility, accounting, performance measurement, or providing troubleshooting aids, etc. It is understood that all such variations are within the scope of the present disclosure. Moreover, not all operations illustrated in a flow logic or method may be required for a novel implementation.

[0181] Where any operation or component discussed herein is implemented in the form of software, any one of a number of programming languages may be employed such as, for example, C, C++, C #, Objective C, Java, Javascript, Perl, PHP, Visual Basic, Python, Ruby, Delphi, Flash, or other programming languages. Software components are stored in a memory and are executable by a processor. In this respect, the term “executable” means a program file that is in a form that can ultimately be run by a processor. Examples of executable programs may be, for example, a compiled program that can be translated into machine code in a format that can be loaded into a random access portion of a memory and run by a processor, source code that may be expressed in proper format such as object code that is capable of being loaded into a random access portion of a memory and executed by a processor, or source code that may be interpreted by another executable program to generate instructions in a random access portion of a memory to be executed by a processor, etc. An executable program may be stored in any portion or component of a memory. In the context of the present disclosure, a “computer- readable medium” can be any medium (e.g., memory) that can contain, store, or maintain the logic or application described herein for use by or in connection with the instruction execution system.

[0182] A memory is defined herein as an article of manufacture and including volatile and/or non-volatile memory, removable and/or non-removable memory, erasable and/or non-erasable memory, writeable and/or re-writeable memory, and so forth. Volatile components are those that do not retain data values upon loss of power. Nonvolatile components are those that retain data upon a loss of power. Thus, a memory may include, for example, random access memory (RAM), read-only memory (ROM), hard disk drives, solid-state drives, USB flash drives, memory cards accessed via a memory card reader, floppy disks accessed via an associated floppy disk drive, optical discs accessed via an optical disc drive, magnetic tapes accessed via an appropriate tape drive, and/or other memory components, or a combination of any two or more of these memory components. In addition, the RAM may include, for example, static random access memory (SRAM), dynamic random access memory (DRAM), or magnetic random access memory (MRAM) and other such devices. The ROM may include, for example, a programmable read-only memory (PROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or other like memory device.

[0183] The devices described herein may include multiple processors and multiple memories that operate in parallel processing circuits, respectively. In such a case, a local interface, such as a communication bus, may facilitate communication between any two of the multiple processors, between any processor and any of the memories, or between any two of the memories, etc. A local interface may include additional systems designed to coordinate this communication, including, for example, performing load balancing. A processor may be of electrical or of some other available construction.

[0184] It should be emphasized that the above-described embodiments of the present disclosure are merely possible examples of implementations set forth for a clear understanding of the principles of the disclosure. It is, of course, not possible to describe every conceivable combination of components and/or methodologies, but one of ordinary skill in the art may recognize that many further combinations and permutations are possible. That is, many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the disclosure. All such modifications and variations are intended to be included herein within the scope of this disclosure and protected by the following claims.

Claims

1. (canceled)
2. A computer-implemented system for managing image data and associated metadata, comprising: an application programming interface (API) resident on a computing device, the API configured to:

receive, via a communications interface, image data and associated metadata from: an image acquisition device; a site comprising one or more image acquisition devices and configured to collect, aggregate, and transmit image data and associated metadata; a storage location storing pre-acquired image data and associated metadata; parse the received image data and associated metadata to identify protected health information (PHI) contained within the metadata; store, via a processor, the identified PHI in a first database, the first database configured to securely store PHI in compliance with one or more privacy regulations; store, via the processor, the image data and a subset of the metadata in a second database, the subset of metadata comprising information necessary for image analysis and expressly excluding PHI and personally identifiable information (PII) unrelated to image analysis; and maintain a data linkage between the entries in the first and second databases via a pseudonymized identifier, wherein the pseudonymized identifier enables authorized correlation between the image data in the second database and the PHI in the first database without storing PHI in the second database.

3. The system of claim 2, wherein the API further comprises a transformation module configured to receive data objects and convert them from a native imaging format to a second, different format, wherein the second format is compliant with a standard used in medical imaging research or clinical workflows.

4. The system of claim 3, wherein the second format is conformant to the Digital Imaging and Communications in Medicine (DICOM) standard.

5. The system of claim 3, wherein the transformation module is further configured to process PHI associated with the data objects, wherein the PHI is included, redacted, or encoded in the second format based on the verified credentials of the requesting user.

6. The system of claim 5, wherein user credentials are authenticated through a role-based access control mechanism, and the level of PHI visibility in the transformed data objects is determined by the assigned role of the user.

7. The system of claim 5, wherein the transformation module logs the applied PHI handling operations along with the user credentials and timestamps for audit purposes.

8. A computer-implemented method for segregating and managing image data and associated metadata, the method comprising: receiving, by an application programming interface (API), image data and associated metadata from: an image acquisition device; a site comprising one or more image acquisition devices and configured to collect, aggregate, and transmit image data and associated metadata; and a storage location storing pre-acquired image data and associated metadata; parsing, by the API, the metadata to identify protected health information (PHI) therein; storing, by the API, the PHI in a first database configured to store PHI in compliance with applicable privacy regulations; storing, by the API, the image data and a subset of the metadata in a second database, the subset of metadata comprising information necessary for image analysis and being devoid of PHI and other personally identifiable information (PII) unrelated to the image data; assigning, by the API, a pseudonymized identifier to link entries between the first and second databases; and enabling, via the API, analytic applications to access the second database and prohibiting access to the first database unless verified user credentials with authorization are provided.

9. An application programming interface (API) system for medical imaging data transformation and protected health information (PHI) management, comprising: a transformation module configured to receive data objects representing medical imaging data in heterogeneous source formats, including native imaging formats and Digital Imaging and Communications in Medicine (DICOM) files that are non-standardized or structurally incommensurate; an ontology-driven mapping engine operatively coupled to the transformation module, the mapping engine configured to apply a domain-specific ontology to govern the transformation of the data objects into a harmonized DICOM-compliant format, wherein the ontology defines standardized mappings, metadata structures, and semantic alignments across data originating from disparate sources to

ensure interoperability among the transformed data objects and compatibility with downstream clinical and research workflows; and a PHI handling module configured to dynamically determine, based on verified credentials of a requesting user, whether PHI associated with the data objects is included, redacted, or encoded within the transformed DICOM-compliant data objects, wherein the PHI handling module applies transformation-time policies based on the user's authenticated role and access privileges.

10. The API system of claim 9, wherein the ontology-driven mapping engine applies standardized DICOM tags, private tags, and structured reporting templates defined by the ontology to ensure semantic consistency among transformed data objects.

11. The API system of claim 10, wherein the ontology comprises mappings that reconcile non-standard or vendor-specific DICOM fields into standardized fields to enable cross-system interoperability.

12. The API system of claim 9, wherein the PHI handling module integrates with a role-based access control system, wherein the user's role determines a PHI visibility policy specifying whether PHI is to be fully included, redacted, or encoded in the transformed data objects.

13. The API system of claim 12, wherein the PHI handling module automatically logs all PHI handling actions performed during transformation, along with user credentials, timestamps, and transformation parameters for auditing and compliance tracking.

14. The API system of claim 9, wherein the transformation module and the PHI handling module operate as part of a unified transformation pipeline to enforce consistent PHI handling during the data object transformation process.

15. A method for transforming medical imaging data and managing protected health information (PHI), the method comprising: receiving, by a transformation module of an application programming interface (API), data objects representing medical imaging data in heterogeneous source formats, including native imaging formats and non-standardized or incommensurate Digital Imaging and Communications in Medicine (DICOM) files; transforming, by the transformation module, the data objects into a harmonized DICOM-compliant format using an ontology-driven mapping engine, wherein the mapping engine applies a domain-specific ontology defining standardized mappings, metadata structures, and semantic alignments to ensure interoperability among the transformed data objects and compatibility with downstream clinical and research workflows; verifying credentials of a requesting user; and dynamically applying, by a PHI handling module, PHI inclusion, redaction, or encoding rules to the transformed DICOM-compliant data objects during transformation, wherein the rules are determined based on the verified user credentials and associated access privileges.

16. The method of claim 15, further comprising applying, by the ontology-driven mapping engine, standardized DICOM tags, private tags, and structured reporting templates defined by the ontology to the transformed data objects.

17. The method of claim 15, further comprising reconciling non-standard or vendor-specific DICOM fields into standardized fields using the mappings defined in the ontology

18. The method of claim 15, wherein verifying the credentials comprises authenticating the user via a role-based access control system, and wherein the applying PHI rules step comprises determining PHI handling policies based on the authenticated user's assigned role.

19. The method of claim 18, further comprising logging, by the PHI handling module, the PHI handling actions performed during the transformation along with the user credentials, timestamps, and transformation parameters for auditing purposes.

20. The method of claim 15, wherein the transforming step and the applying PHI rules step are performed within a unified transformation pipeline to enforce consistent PHI handling during data object transformation.
